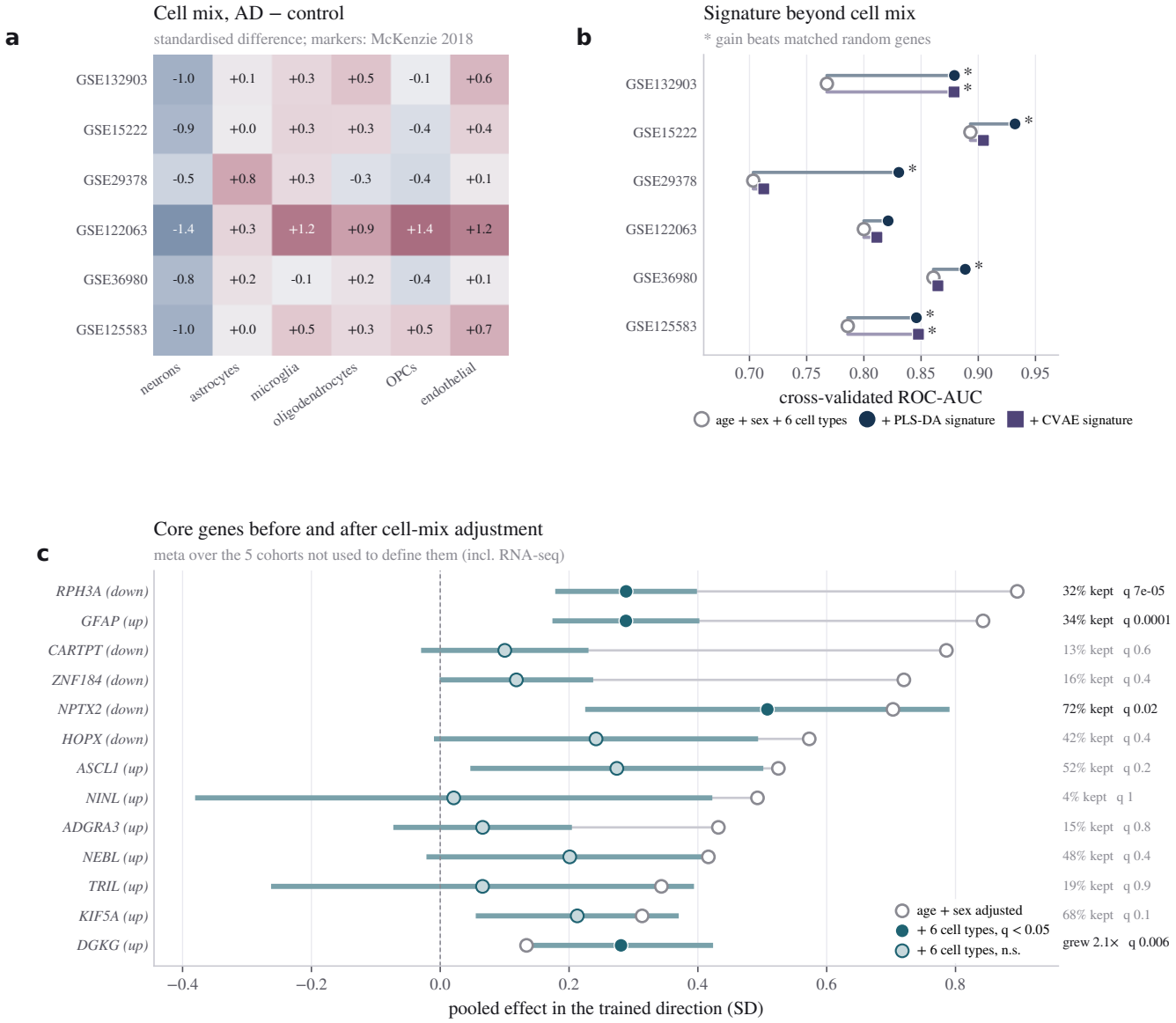
a
Cell mix, AD − control
standardised difference; markers: McKenzie 2018
GSE132903 -1.0 +0.1 +0.3 +0.5 -0.1 +0.6
GSE15222 -0.9 +0.0 +0.3 +0.3 -0.4 +0.4
GSE29378 -0.5 +0.8 +0.3 -0.3 -0.4 +0.1
GSE122063 -1.4 +0.3 +1.2 +0.9 +1.4 +1.2
GSE36980 -0.8 +0.2 -0.1 +0.2 -0.4 +0.1
GSE125583 -1.0 +0.0 +0.5 +0.3 +0.5 +0.7
neurons
astrocytes
microglia
oligodendrocytes
OPCs
endothelial
b
Signature beyond cell mix
* gain beats matched random genes
GSE132903
GSE15222
GSE29378
GSE122063
GSE36980
GSE125583
0.70 0.75 0.80 0.85 0.90 0.95
cross-validated ROC-AUC
age + sex + 6 cell types
+ PLS-DA signature
+ CVAE signature
c
Core genes before and after cell-mix adjustment
meta over the 5 cohorts not used to define them (incl. RNA-seq)
RPH3A (down) 32% kept q 7e-05
GFAP (up) 34% kept q 0.0001
CARTPT (down) 13% kept q 0.6
ZNF184 (down) 16% kept q 0.4
NPTX2 (down) 72% kept q 0.02
HOPX (down) 42% kept q 0.4
ASCL1 (up) 52% kept q 0.2
NINL (up) 4% kept q 1
ADGRA3 (up) 15% kept q 0.8
NEBL (up) 48% kept q 0.4
TRIL (up) 19% kept q 0.9
KIF5A (up) 68% kept q 0.1
DGKG (up) grew 2.1× q 0.006
age + sex adjusted
+ 6 cell types, q < 0.05
+ 6 cell types, n.s.
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8
pooled effect in the trained direction (SD)